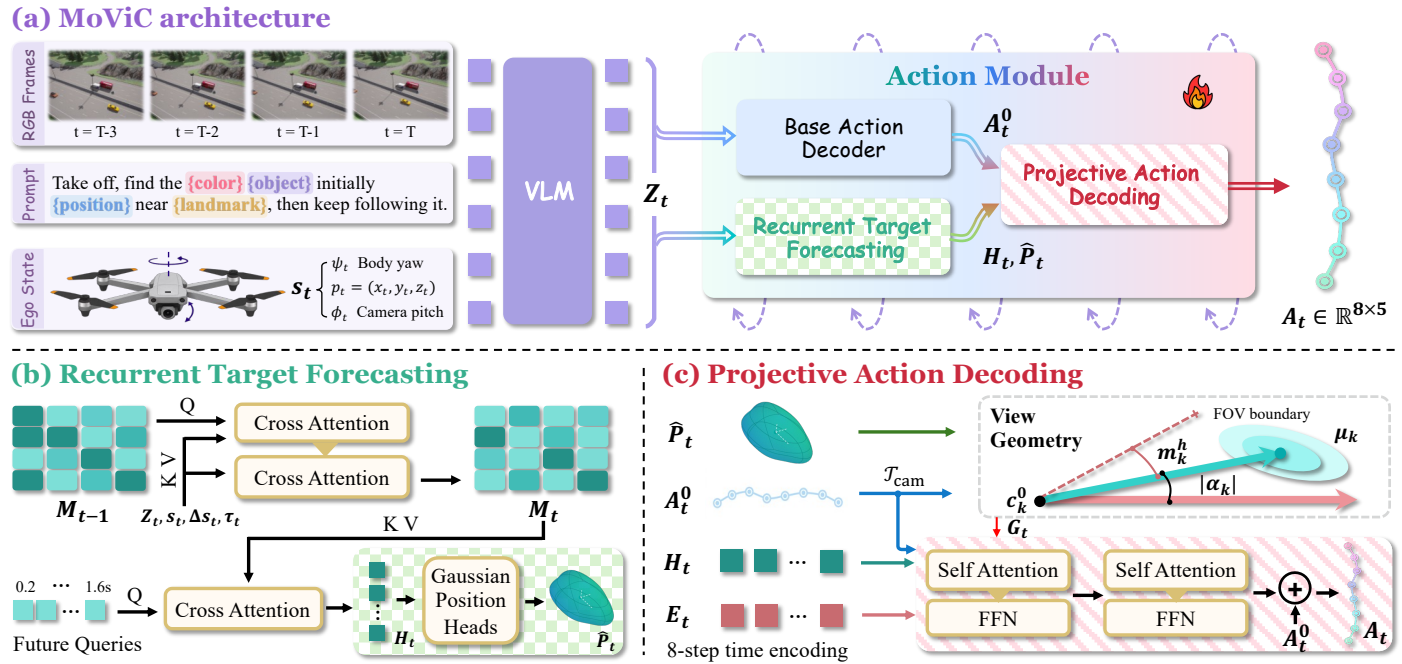

(a) MoViC architecture
RGB Frames
t = T-3
t = T-2
t = T-1
t = T
Prompt
Take off, find the {color} {object} initially {position} near {landmark}, then keep following it.
Ego State
s_t
ψ_t Body yaw
$p_t = (x_t, y_t, z_t)$
ϕ_t Camera pitch
VLM
Z_t
Action Module
Base Action Decoder
A_t^0
Projective Action Decoding
Recurrent Target Forecasting
$H_t, \hat{P}_t$
$A_t \in \mathbb{R}^{8\times5}$
(b) Recurrent Target Forecasting
M_{t-1}
Q
K V
Cross Attention
Cross Attention
$Z_t, s_t, \Delta s_t, \tau_t$
M_t
K V
0.2 ··· 1.6s
Q
Cross Attention
Future Queries
H_t
Gaussian Position Heads
$\hat{P}_t$
(c) Projective Action Decoding
$\hat{P}_t$
A_t^0
H_t
E_t
8-step time encoding
$\mathcal{T}_{\text{cam}}$
View Geometry
FOV boundary
m_k^h
μ_k
$|\alpha_k|$
c_k^0
G_t
Self Attention
FFN
Self Attention
FFN
A_t^0
A_t